

Junghyun Min

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SKILLS & SERVICE

Software Engineering: Unix, databases, Flask, FastAPI, asynchrony, Google Cloud, Docker, Portainer, Git, Jenkins.

Natural Language Processing: PyTorch, TensorFlow, transformers, TorchServe, scikit-learn, ML baselines

Large Language Models: OpenAI, LangChain, RAG, prompt engineering. vLLM, quantization, distributed training

Languages: Python, Java, C++, JavaScript, R, Wolfram; English/Korean (Native), German/Mandarin (Working)

Service: Reviewer for *ACL, Language Resources and Evaluation, Machine Learning Engineering; NACLO Site Host

EDUCATION

Georgetown University

Ph.D. Computer Science and Linguistics. Advisor: Ethan Wilcox.

Washington, DC

Exp. 2029

Johns Hopkins University

M.A. Cognitive Science. Advisor: Tal Linzen.

Baltimore, MD

Dec 2020

B.S. Physics, B.A. Mathematics. General Honors, early graduation.

Dec 2017

WORK EXPERIENCE

Nokia Bell Labs

Machine Learning and AI Intern

Murray Hill, NJ

Jun 2026 – Aug 2026

- Designed and implemented an LLM-based agentic network traffic simulation workflow for synthetic data generation

University of Toronto

Computer Science Research Intern

Toronto, ON

May 2025 – Aug 2025

- Pre-trained LMs from scratch and via transfer; compiled 7-task Cantonese NLU benchmark and established baselines
- Performed large-scale verification of a novel training objective across architecture, language, and resource settings

NCSOFT

Machine Learning Engineer (Natural Language Processing) at Language AI Lab

Seongnam, Korea

Jan 2021 – Apr 2024

- Implemented and trained tokenizers for the Language Model Task Force, precursor to the 1.3-13B [VARCO LLMs](#)
- Deployed efficient, scalable Stanza-like cloud REST API, parsing 10k requests per sec on 4GB VRAM at 96% acc.
- Served machine learning models for information retrieval via Docker and TorchServe, processing 30+ sent per sec.
- Designed algorithms in deep learning models to improve information extraction performance (+30%p) and stability
- Built and optimized agentic pipelines to automatically parse current news to extract and rank 1,000 events per day

SELECTED TECHNICAL PROJECTS

Lead engineer, Agentic document automation platform [genDOC](#)

- Built RAG-based agentic document automation software with LangChain and FastAPI on OpenAI API using Python
- Implemented orchestration framework, routing pipeline, agent-wise output verification logic and self-correction loops
- Optimized RAG pipeline to achieve 90+% end-to-end pass rate (pass@5) and end-to-end latency under 60 seconds

Lead software engineer, ai.ly

- Fine-tuned and cloud deployed a GPT-based AI lyricist that generates lyrics aligned to user preferences on PyTorch
- Accumulated 50k+ visits over 3 months of service, cross-functional collaboration resulting in a [hip-hop sample](#)

Co-lead, [Large language models and legal interpretation](#)

- Implemented LLM legal interp. judgment extraction tool with vLLM. [Oral presentation](#), NLLPW at EMNLP 2025
- Showed even large (70B) and commercial LLMs are sensitive to surface form, lack alignment to human judgment

Lead, Multimodal text-prosody model for syntactic information content estimation

- Developed and fine-tuned 3-modal transformer architecture for syntax prediction with multimodal text and audio input
- Measured syntax information content in text (-74%), duration (-3%), pause (-1.7%) in naturalistic and planned speech

SELECTED PUBLICATIONS

- Junghyun Min**, Minhoo Lee, Woohul Lee, Yeonsoo Lee. RepL4NLP at NAACL 2025. [Punctuation Restoration Improves Structure Understanding without Supervision](#). [Tech blog \(Korean\)](#). PyTorch implementation on Docker.
- Abhishek Purushothama, **Junghyun Min**, Brandon Waldon, Nathan Schneider. FAccT 2026. [Prompting from the bench: Large-scale pretraining is not sufficient to prepare LLMs for ordinary meaning analysis](#). vLLM and OpenAI API.
- Junghyun Min**, R. Thomas McCoy, Dipanjan Das, Emily Pitler, and Tal Linzen. ACL 2020. [Syntactic data augmentation increases robustness to inference heuristics](#). Collaboration with Google Research using TensorFlow.